

BYUNGJOO CHAE

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EDUCATION

M.S. Electronic Engineering (Advisor: Donghyeon Cho)

Mar 2022 - Feb 2024

Chungnam National University, Daejeon, Republic of Korea

B.S Electronic Engineering

Mar 2016 - Feb 2022

Chungnam National University, Daejeon, Republic of Korea

WORK EXPERIENCE

Machine Learning Engineer, AIPIA

Feb 2025 -

- Poptale

- Built a preprocessing pipeline using Airflow to automate and reliably manage the text-to-image data generation process

Machine Learning Engineer, Dexter Studios

Mar 2024 - Jan 2025

- De-aging Pipeline for East Asian VFX Productions

- Developed a de-aging model with an East Asian face dataset and developed a **frame-consistent video pipeline used in commercial VFX workflows**.
- Fine-tuned InstructPix2Pix using a curated de-aging dataset, achieving **90% faster inference time** for internal testing and experimentation.

Researcher, Chungnam National University

Mar 2022 - Feb 2024

- Patch-based Painterly Harmonization for High Resolution Images

- Built **10,000+ high-resolution images** for painterly harmonization using Ultra Style Transfer models.
- Developed a patch-based harmonization method that combined local and global features, leading to a **+0.3 PSNR and 10 MSE improvement** over the base model.

- Deep Real: Synthetic Data Generation and Model Fine-Tuning

- Designed and implemented a pipeline to automatically **generate 26,157 synthetic harmonization dataset** and fine-tuned the HRNet-IDIH model to improve harmonization performance.
- Evaluated the fine-tuned model on 1,000 images, **achieving a 5-point increase in PSNR** compared to the base model

PERSONAL PROJECTS

Financial Agent AI , PseudoLab

Mar 2025 -

- Developed LLM-based agents using LangChain and LangGraph, including a Plan Agent and meme recommendation Agent, for a financial multi-agent chatbot system.
- Built an Airflow-based ETL pipeline and managed data storage with PostgreSQL to support meme stock recommendation agent AI.

PUBLICATION

- Online Learning for Reference-Based Super-Resolution, **Byungjoo Chae***, Jinsun Park*, Tae-Hyun Kim and Donghyeon Cho, *MDPI Electronics*, 2022.
- Learning Lightweight Low-Light Enhancement Network using Pseudo Well-Exposed Images, Seonggwon Ko*, Jinsun Park*, **Byungjoo Chae** and Donghyeon Cho, *IEEE Signal Processing Letter (SPL)*, 2021

TECHNICAL SKILLS

- Advanced: Python, Pytorch, Intermediate: Docker, Github, Beginner: FastAPI, langchain, langgraph, Airflow